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機器心智理論推理：訓練、測試、與應用

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11/06/2026 @ 長庚大學

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- **國立臺灣大學**資訊工程學系 助理教授（2025 至今）
- **本田美國研究院** AI科學家（2023 – 2025）
- **亞馬遜公司** 實習研究員（2021、2022 暑假）

- **約翰霍普金斯大學**電機與電腦工程 博士（2019 – 2023）
- **國立交通大學**電子研究所 碩士（2017 – 2019）
- **國立交通大學**電機資訊學士班 學士（2013 – 2017）

Outline

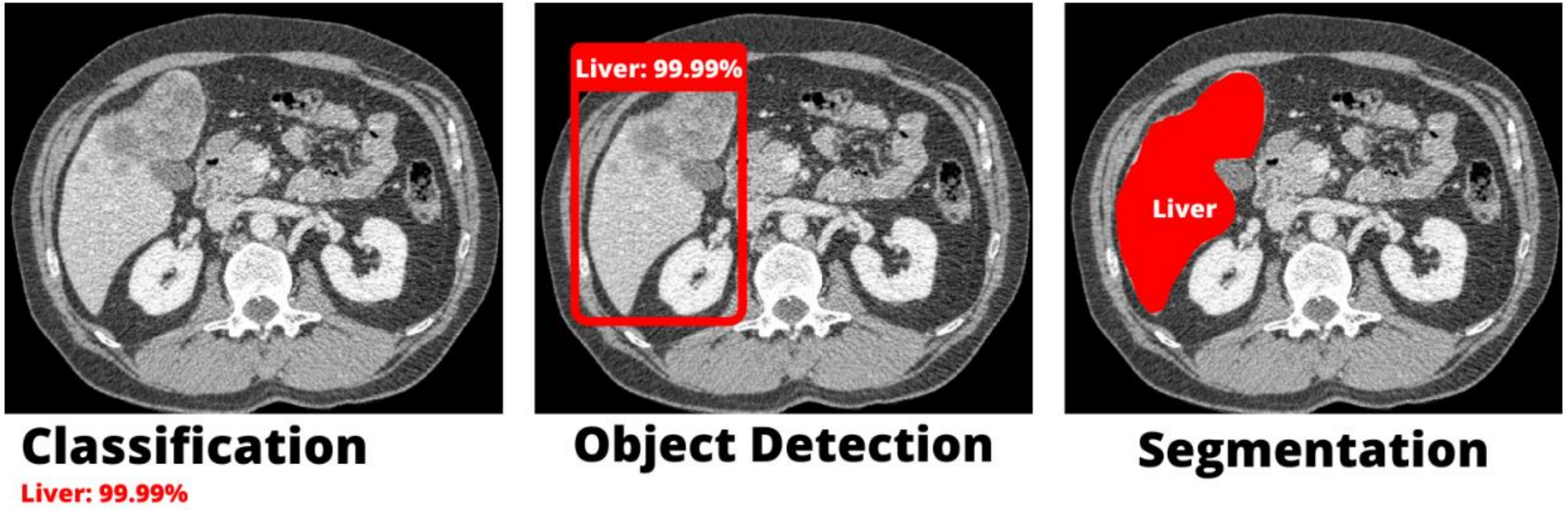
- 大型推理語言模型的訓練三步驟

- 機器心智理論推理：訓練、測試、與應用

大型推理語言模型的訓練三步驟

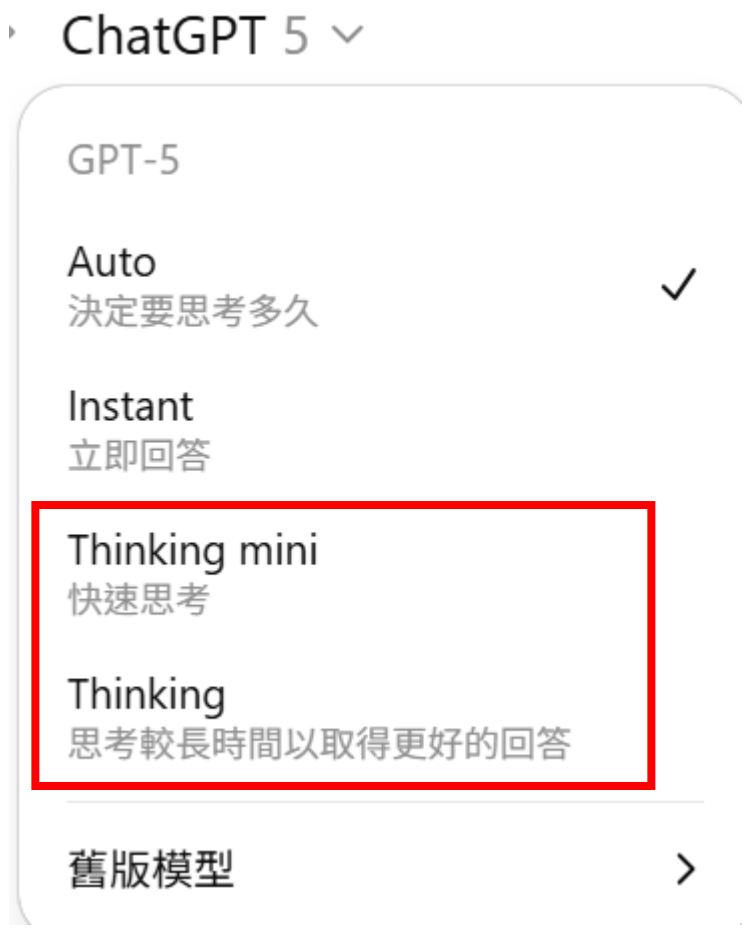
What Can Be Done Without LLMs?

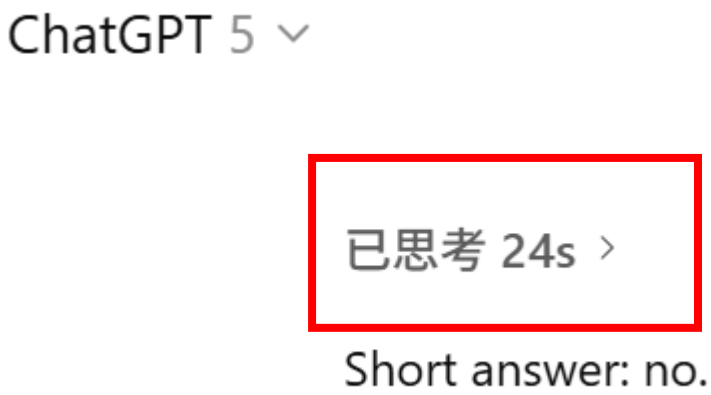
- Input → Output (input-output mapping)
- Task-specific perception, fixed output format, not explainable



[Source credit: https://github.com/aminbenmansour/automatic-liver-segmentation]

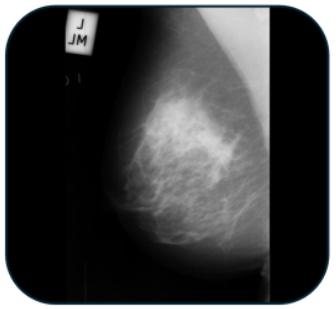
What Are Reasoning LLMs?





What Reasoning LLMs Enable?

- Input → Reasoning → Output (thinking to answer)
- Flexible tasks, open-ended, natural language interaction, explainable



Problem: What does this image depict in terms of its content?

A)Brain tissue
B)Intestinal tissue
C)Kidney tissue
D)Breast tissue

Med-R1 Response:

Think: The image appears to be a mammogram, which is a type of X-ray used to detect abnormalities in the breast tissue. The presence of dense, irregular patterns and a central area of lighter density suggest the presence of breast tissue.

Answer: D

Groundtruth: D



Problem: What is the specific abnormality present in this image?

A)Pneumonia
B)Sinusitis
C)Urolithiasis
D)Osteoporosis

Med-R1 Response :

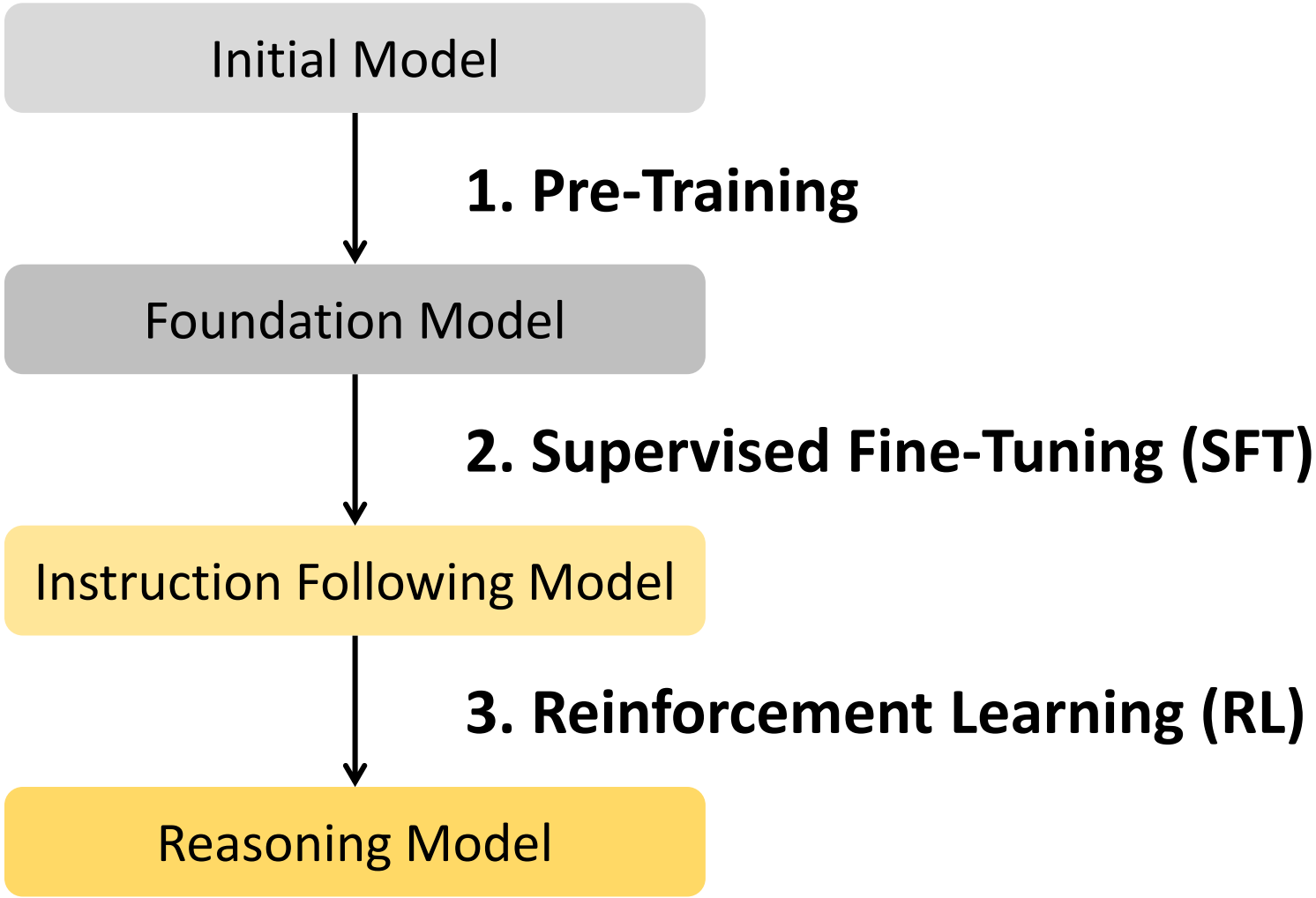
Think: The image is a computed tomography (CT) scan of the abdomen, which is a medical imaging technique used to visualize internal structures. The specific abnormality present in this image is not immediately clear from the visible structures alone. However, the presence of a large, dense mass in the right lower quadrant of the abdomen could suggest a urolithiasis.

Answer: C

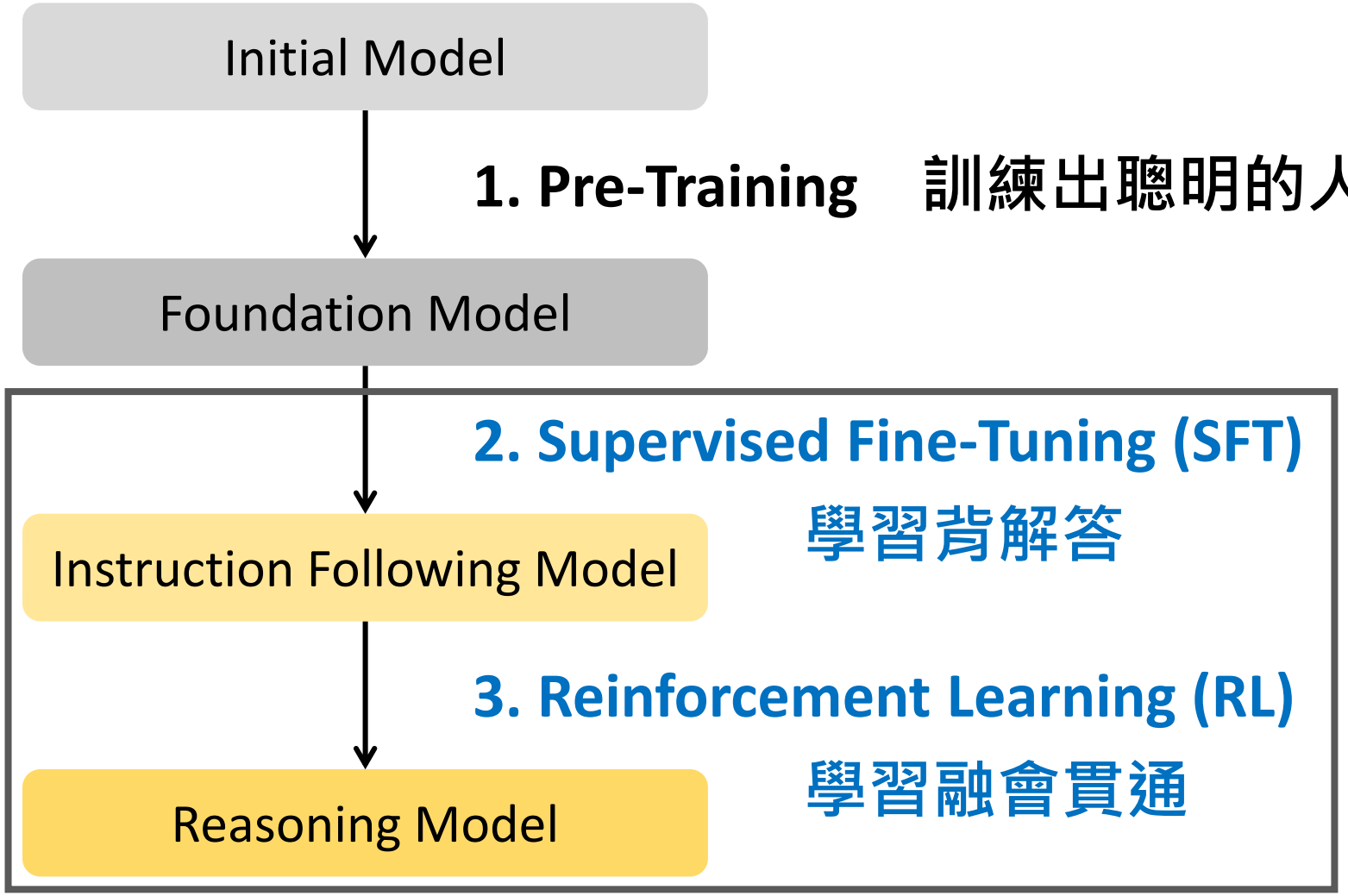
Groundtruth: C

[Lai et al. 2025]

So, How to Train a Reasoning LLM? 3 Stages!



So, How to Train a Reasoning LLM? 3 Stages!



Post-Training

將人才訓練成專家

1. Pre-Training

- Next-token prediction 文字接龍

$$P_\theta(x_t \mid x_{<t})$$

where $x_{<t} = (x_1, \ldots, x_{t-1})$

Training objective:

$$P_\theta(x_1, \ldots, x_T) = \prod_{t=1}^{T} P_\theta(x_t \mid x_{<t})$$

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我來_
我來到_
我來到長_
我來到長庚_
我來到長庚大_
我來到長庚大學_
我來到長庚大學人_
我來到長庚大學人工_
我來到長庚大學人工智_
我來到長庚大學人工智慧_
我來到長庚大學人工智慧學_

1. Pre-Training

- Formulates all types of knowledge as a next-token prediction problem (autoregressive modeling)

Task	**Example**
Grammar	In my free time, I like to __. {**run**, banana}
Lexical semantics	I spread butter on warm __. {**bread**, pencil}
World knowledge	The tallest mountain in the world is __. {**Mount Everest**, Mount Fuji}.
Sentiment analysis	Restaurant review: The food was delicious and the service was friendly. The experience was __. {**positive**, negative}
Translation	The word for “cat” in French is __. {**chat**, perro}
Spatial reasoning	The ball rolled under the __. {**table**, sky}
Logical reasoning	If it rains, the ground gets wet. If the ground is wet, the game is canceled. It rained today, so the game is __. {**canceled**, continued}
Math	Second grade arithmetic exam: 7 × 6 = __. {**42**, 36}
Coding	def factorial(n): return 1 if n == 0 else n * __. {**factorial(n-1)**, factorial(n+1)}

1. Pre-Training

- Example: English-French translation
- The knowledge **naturally occurs** in **web-scale** corpora
- Uses **next-token prediction** to learn the knowledge from web-scale corpora

”I’m not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbecile [I’m not a fool].**

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: ”**Mentez mentez, il en restera toujours quelque chose**,” which translates as, ”**Lie lie and something will always remain.**”

“I hate the word ‘**perfume**,’” Burr says. ‘It’s somewhat better in French: ‘**parfum**.’

If listened carefully at 29:55, a conversation can be heard between two guys in French: “**-Comment on fait pour aller de l’autre coté? -Quel autre coté?**”, which means “**- How do you get to the other side? - What side?**”.

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

“**Brevet Sans Garantie Du Gouvernement**”, translated to English: “**Patented without government warranty**”.

[Radford et al. 2019]

1. Pre-Training

- Next-token prediction **doesn't need anything else** but corpora
- Less requirements means dealing with more data: ebook, code, html…

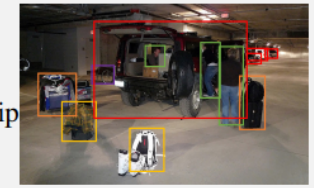
- No human annotations → **just raw text**
- No format requirement → **just raw text**
- No structure parsing → **just raw text**

- **Scalability!** → **Web-scale training data**

2. Supervised Fine-Tuning (SFT)

- Learns specified knowledge (to be an expert) from **carefully designed and annotated data** (supervised)
- The annotations demonstrate how human responds to an instruction
- Collecting annotations is expensive (by human and/or GPT)

Context type 1: Captions
A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes
person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation
Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>
Response type 2: detailed description
The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>
Response type 3: complex reasoning
Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

[Liu et al. NeurIPS'23]

2. Supervised Fine-Tuning (SFT)

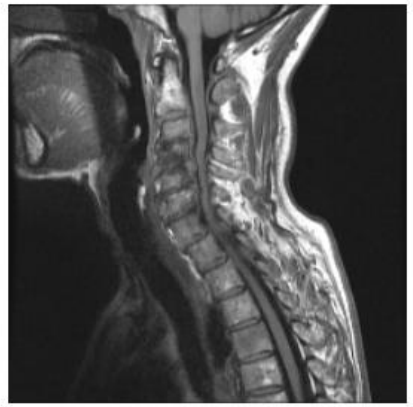
- Uses next-token prediction to learn as well
- 背解答模板

$$\text{Input} \rightarrow \underbrace{\text{Step}_1 \rightarrow \text{Step}_2 \rightarrow \cdots \rightarrow \text{Step}_n}_{\text{Reasoning chain}} \rightarrow \text{Answer}$$

SFT data

Question: What is the finding in this image?

A) Stomach ulcer, B) Nerve compression,
C) Cord pathology, D) Liver cirrhosis



Answer: To determine the correct answer, we need to identify the finding in the image provided. The MRI scan in question is likely a view of a spine, not the stomach, or the liver. Stomach ulcers, liver cirrhosis, and spinal cord pathology can be diagnosed with MRI, but the image in question shows a spinal view, not a gastrointestinal or abdominal view. Therefore, the only reasonable options left are nerve compression and cord pathology.

C

[Pan et al. MICCAI'25]

3. Reinforcement Learning (RL)

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融會貫通

- Learns to reason by itself → better generalization
- The RL reward evaluates the quality of the final answer while leaving the reasoning process unconstrained

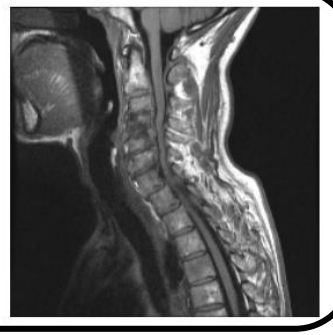
$$\boxed{\text{Input}} \rightarrow \underbrace{\text{Step}_1 \rightarrow \text{Step}_2 \rightarrow \cdots \rightarrow \text{Step}_n}_{\text{Reasoning chain}} \rightarrow \boxed{\text{Answer}}$$

RL data

- RL training data require only the final answer to be annotated → much less expensive than SFT data

Question: What is the finding in this image?

A) Stomach ulcer, B) Nerve compression,
C) Cord pathology, D) Liver cirrhosis

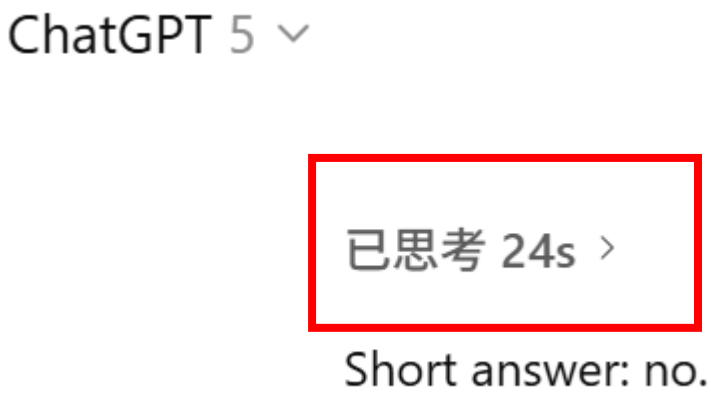


Answer: C

[Pan et al. MICCAI'25]

3. Reinforcement Learning (RL)





Comparison

Stage	Goal	Data	Data cost	Data amount	Learning signal
1. Pre-Training	Learns language modeling and generic knowledge	Massive unlabeled corpus	Low	Web-scale	Next-token prediction
2. SFT	Learns specified knowledge to be an expert	High-quality labeled instructions	Very high	Thousands to millions	Next-token prediction
3. RL	Learns to reason by itself for better generalization	Labeled final answers	Medium	Thousands to millions	Reward score

So, How to Train Our Own Expert LLM?

- Pre-training is crazy expensive 
- We can only handle the cost of **post-training** ☺

1. Starts from an open-source pre-trained model (Qwen, DeepSeek, etc.)
2. Collects and annotates our own SFT data and RL data for our tasks
3. Performs SFT and RL using our training data

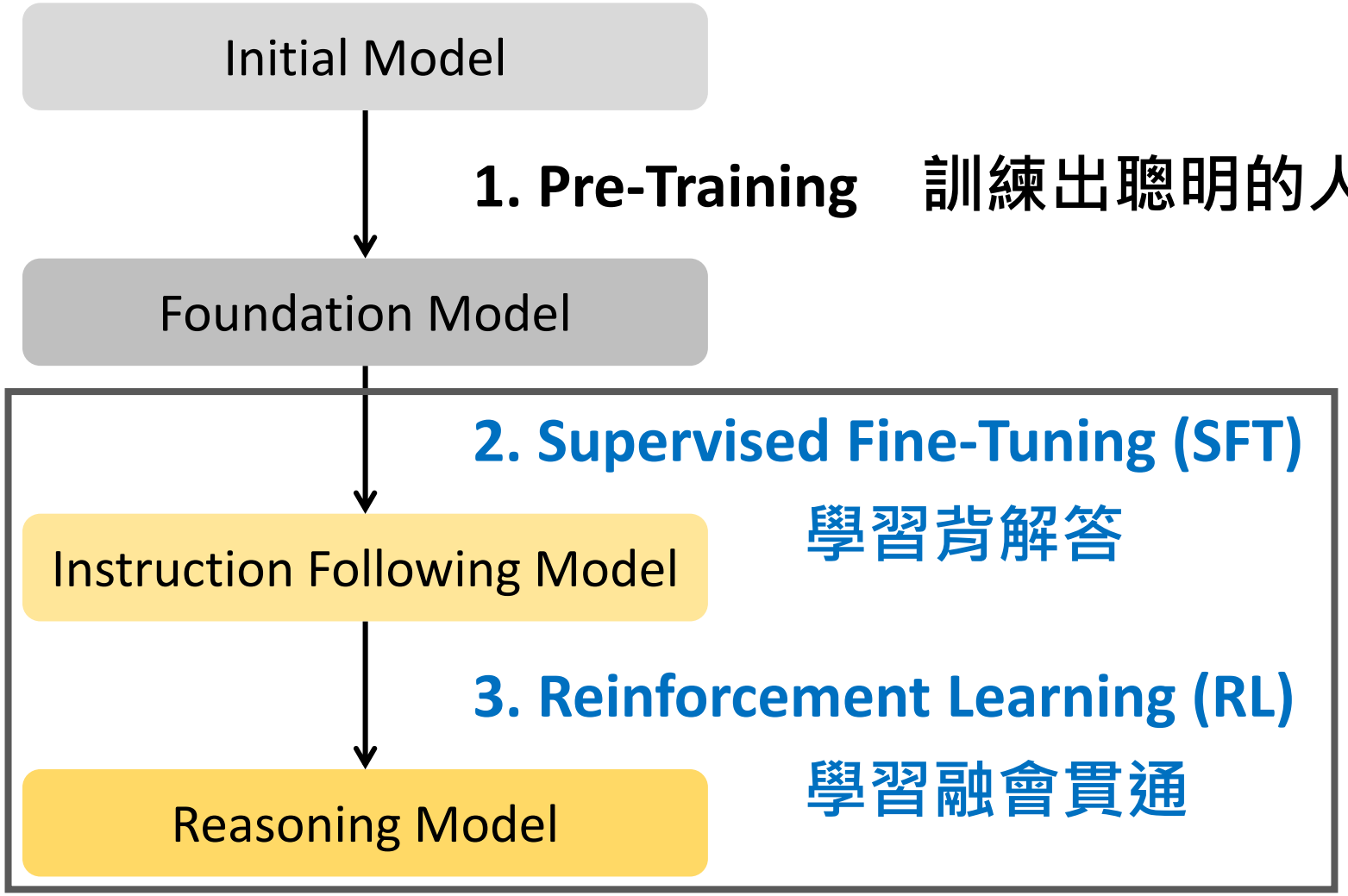
$$\text{Input} \rightarrow \underbrace{\text{Step}_1 \rightarrow \text{Step}_2 \rightarrow \cdots \rightarrow \text{Step}_n}_{\text{Reasoning chain}} \rightarrow \text{Answer}$$

SFT data

$$\boxed{\text{Input}} \rightarrow \underbrace{\text{Step}_1 \rightarrow \text{Step}_2 \rightarrow \cdots \rightarrow \text{Step}_n}_{\text{Reasoning chain}} \rightarrow \boxed{\text{Answer}}$$

RL data

Summary



Post-Training

將人才訓練成專家

機器心智理論推理：訓練、測試、與應用

What is Theory of Mind (ToM)?

- ToM is the ability to **understand other people's mental states**, such as thoughts, emotions, intentions, and beliefs
- **Machine ToM** aims to replicate this human's innate ability in AI agents



[He et al. EMNLP-Findings'23]



ToM Promotes Effective Human–AI Collaboration

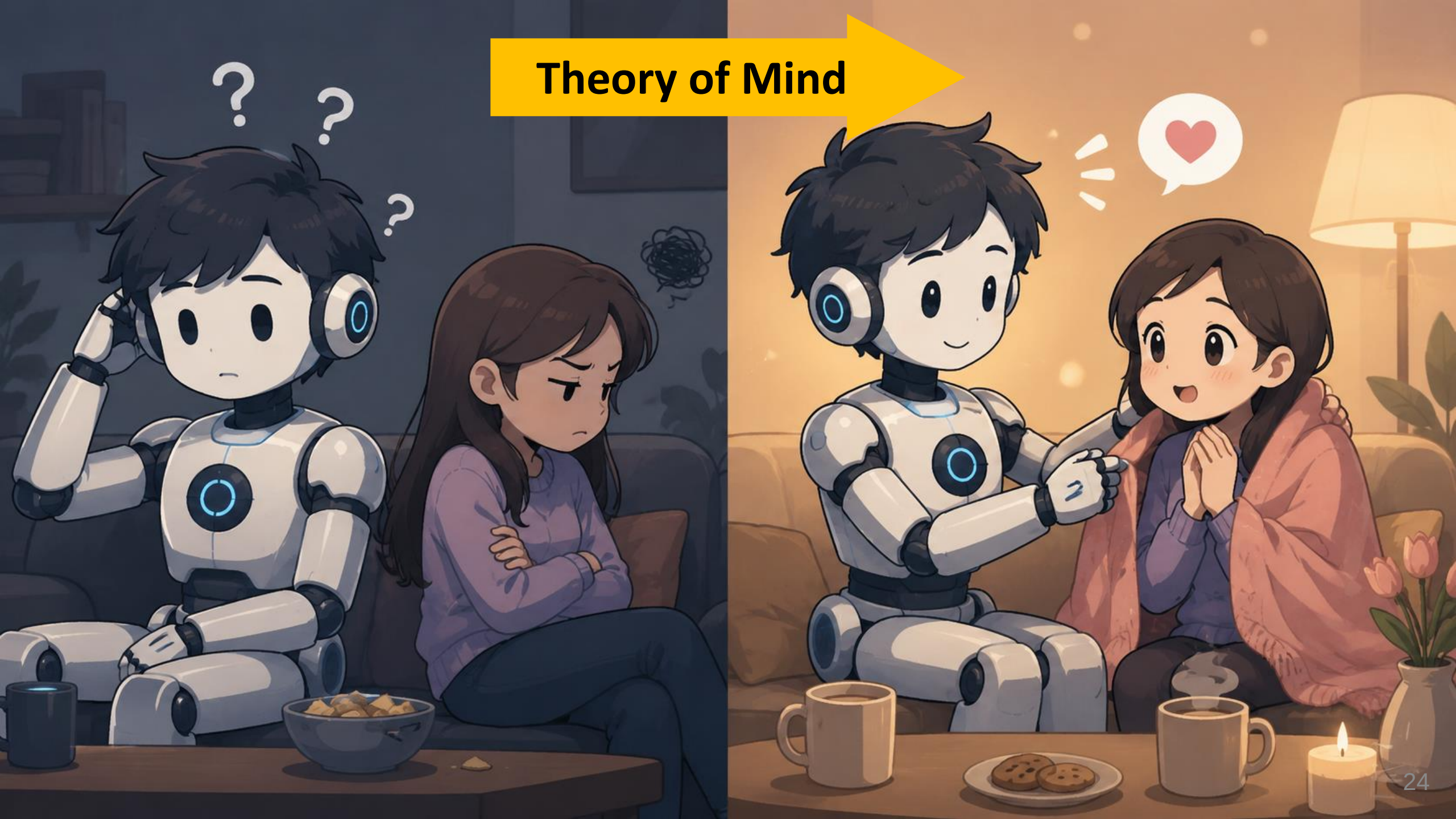
- Infer user's mental states, such as thoughts, intentions, and beliefs
- Track perspectives (what the human knows vs. doesn't know)
- Predict next actions and anticipate needs

→ More aligned assistance

→ Fewer misunderstandings

→ Safer interactions




Theory of Mind

Machine ToM Across AI Development

Training



Cost-efficient training
(ICML 2025 Spotlight)

Evaluation



Robust evaluation
(ICML 2026)

Applications



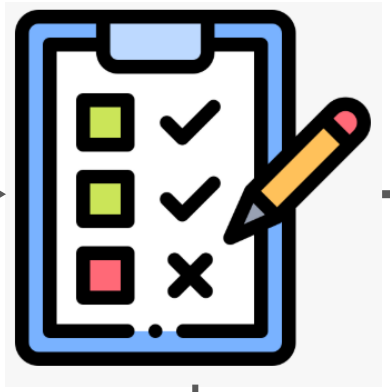
- **Mental manipulation analysis**
- **Social deduction game**
- **Autonomous driving**

Machine ToM Across AI Development

Training



Evaluation



Applications



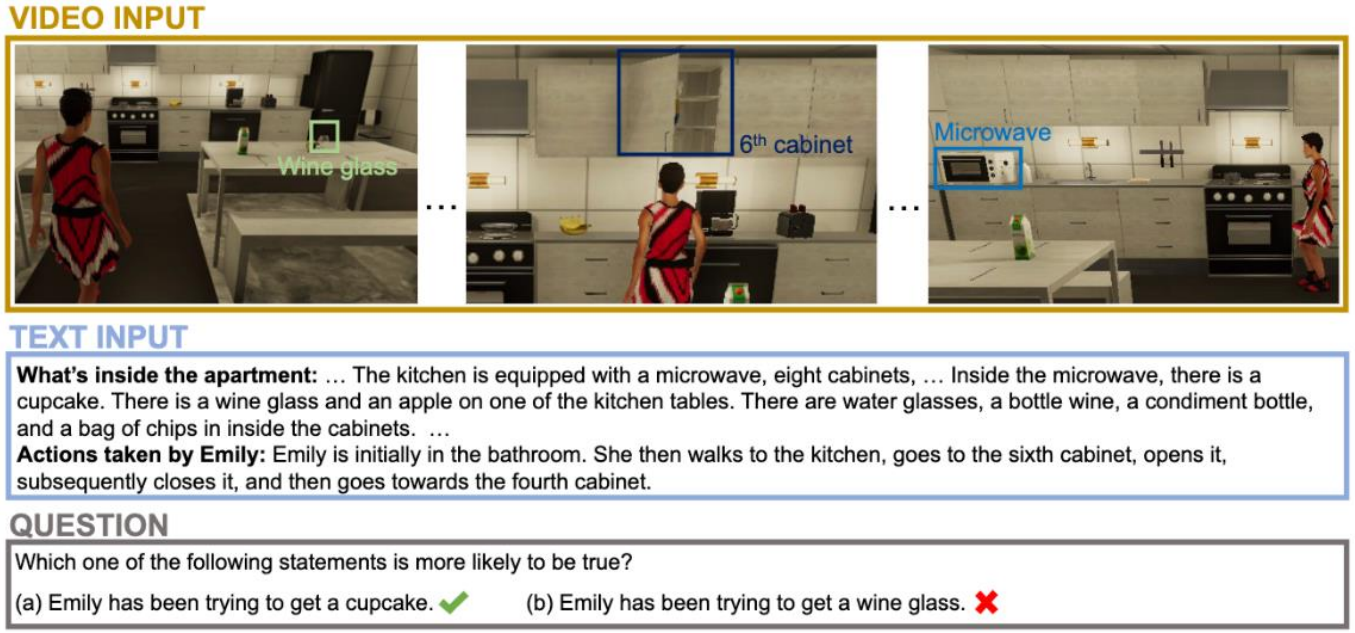
Cost-efficient training
(ICML 2025 Spotlight)

Robust evaluation
(ICML 2026)

- **Mental manipulation analysis**
- **Social deduction game**
- **Autonomous driving**

Multimodal ToM Training is Expensive

- Scaling Law: Larger models generally exhibit stronger capabilities
- However, multimodal ToM training is expensive
- How can we efficiently scale multimodal ToM training to larger models, such as Llama3.1-**405B**?



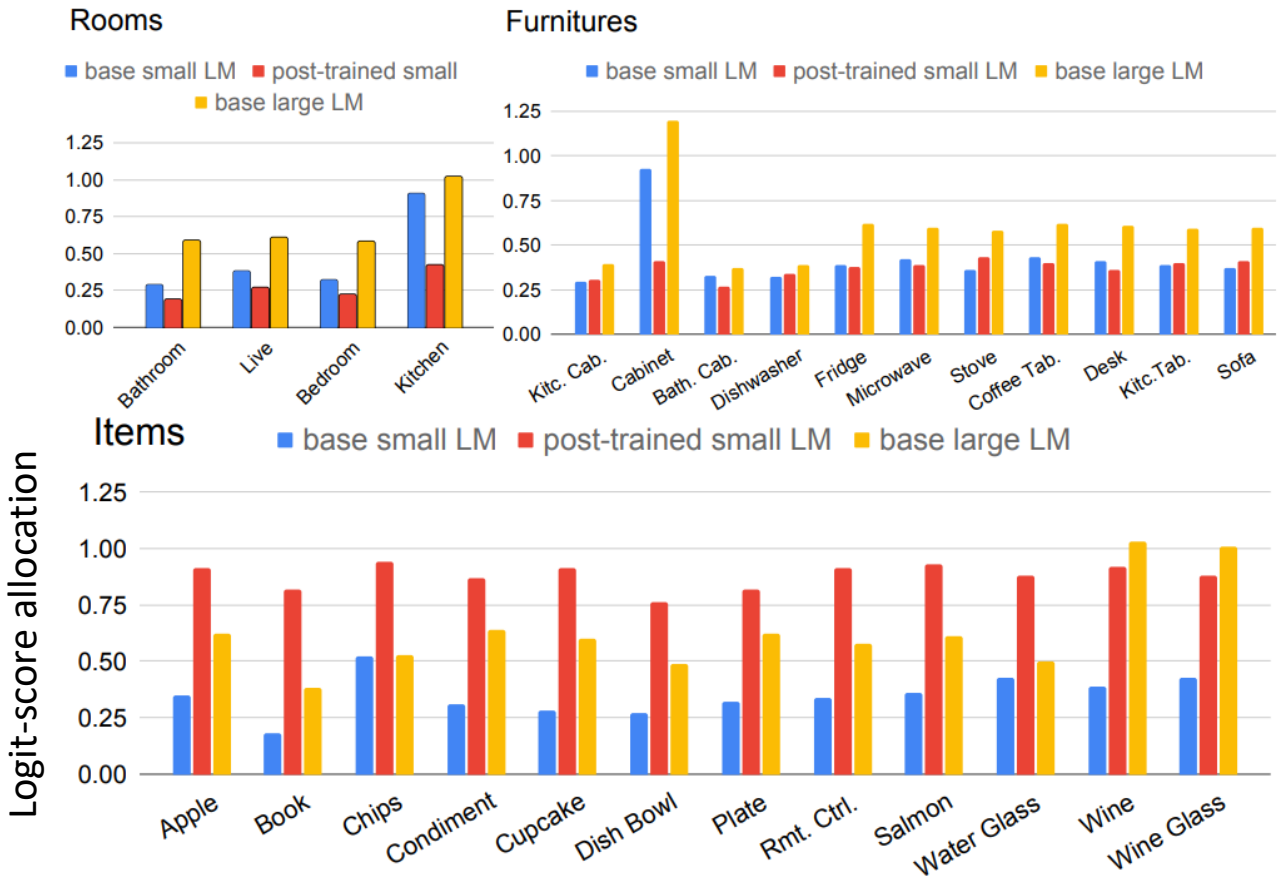
[Jin et al. ACL'24]

Model Behavior Analysis

- **Base Small LM** (8B)
  Post-Trained Small LM (8B)
  Base Large LM (405B)

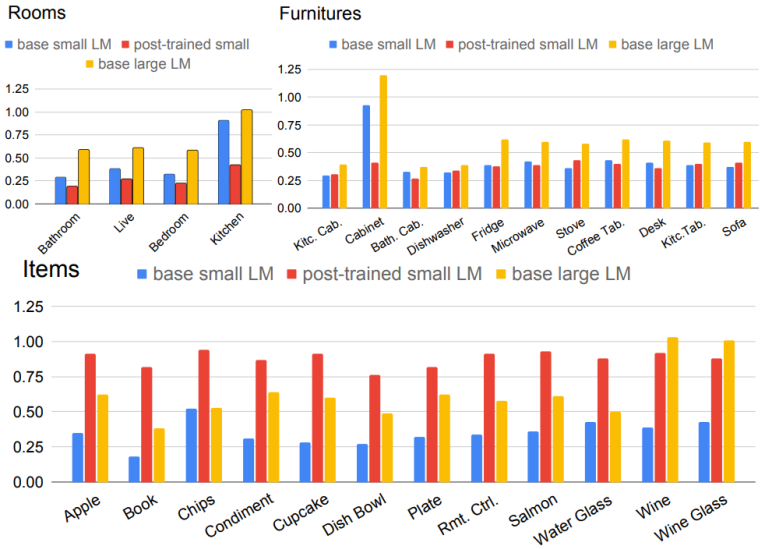
- Three levels of concept granularity:
  Room
  └ **Furniture**
  └ **Item**

How much does the model rely on each concept level during decision-making?



Model Behavior Analysis

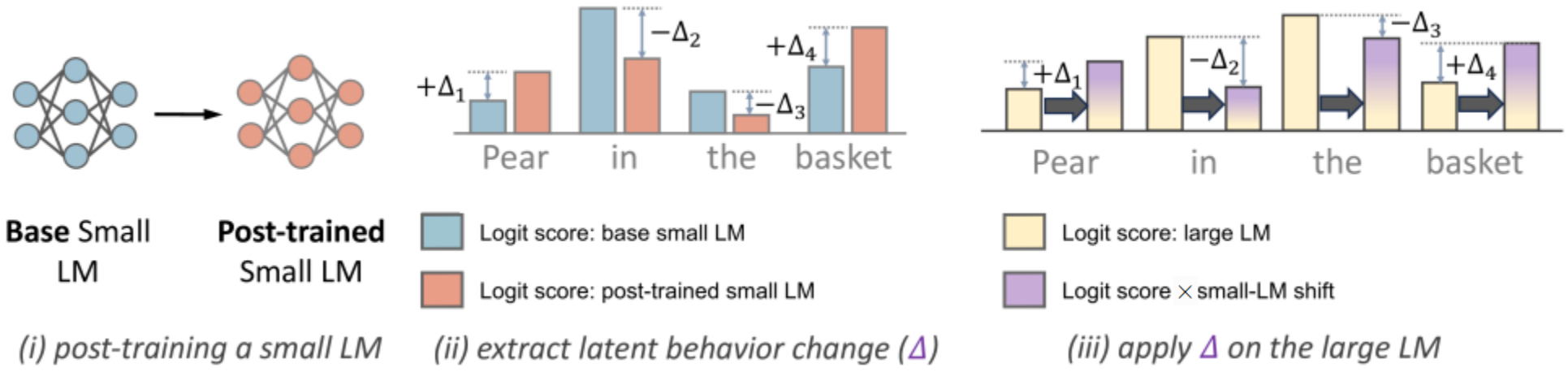
- **Post-Trained Small LM** and **Base Large LM** exhibit distinct behaviors and complementary strengths



Model	Concept level	General
Post-Trained Small LM	Concentrate on fine-grained objects (item)	Align better with specific ToM scenarios
Base Large LM	Reason over higher-level concepts (room, furniture)	Have better general world knowledge and reasoning

Weak-to-Strong Control Approach

- Transfer the post-trained alignment from a **Small LM** to a **Large LM**
- Adapt the **Large LM**'s ToM behavior by training only the **Small LM**



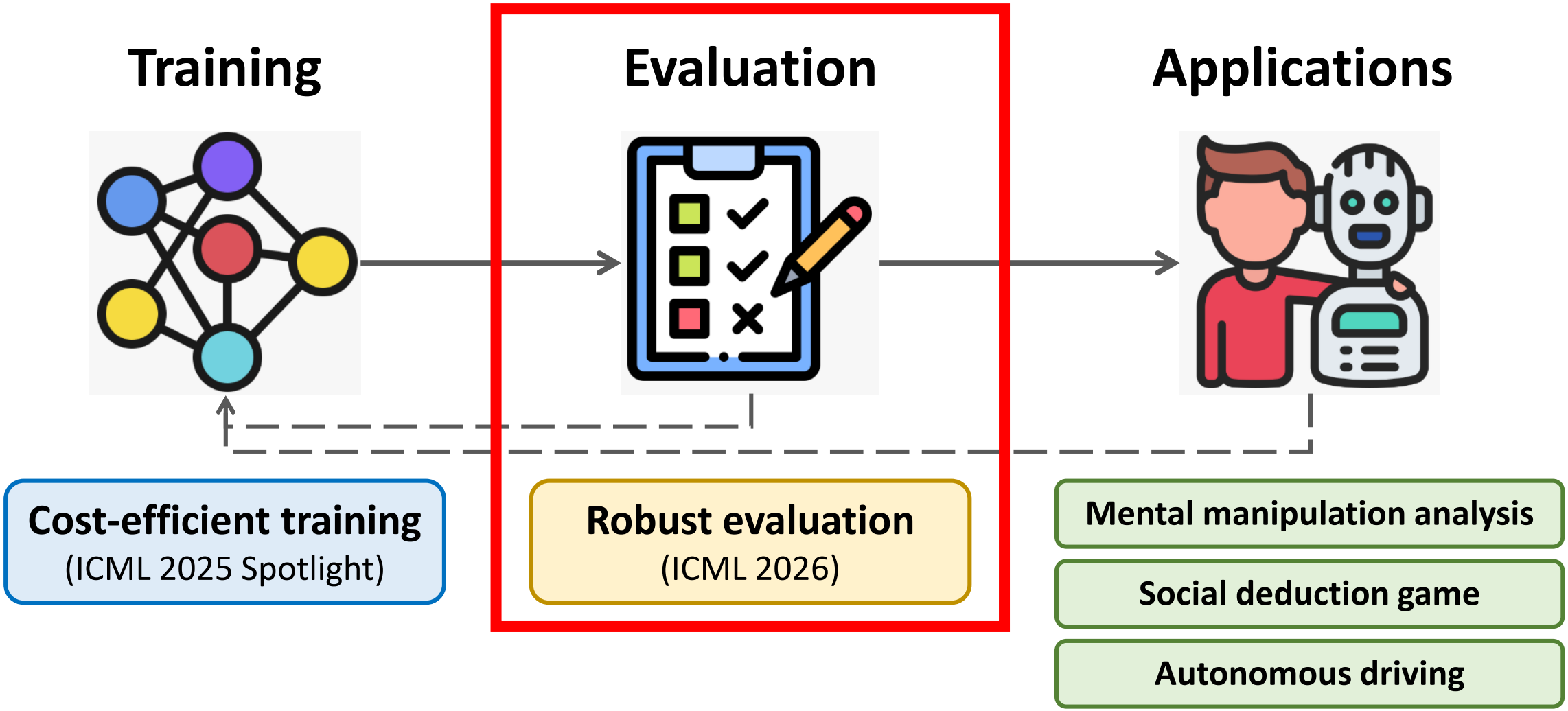
$$\text{Logits}_{\text{adapted large}} = \text{Logits}_{\text{base large}} \times \left(\frac{\text{Logits}_{\text{post-trained small}}}{\text{Logits}_{\text{base small}}} \right)$$

Results

- **Dataset**: MMToM-QA
- **Metric**: Accuracy
- **Ours**: post-train Llama-3.1-8B + weak-to-strong control

LM	config	belief inference				goal inference					all
		1.1	1.2	1.3	avg.	2.1	2.2	2.3	2.4	avg.	
Llama-3.1	8B-zero-shot	88.00	72.00	91.00	83.67	65.33	62.67	22.67	54.67	51.33	65.19
	8B-post-trained	90.00	71.00	93.00	84.67	69.33	72.00	62.67	72.00	69.00	75.71
	70B-zero-shot	85.00	63.00	93.00	80.33	72.00	76.00	16.00	61.33	56.33	66.62
	70B-post-trained	91.00	69.00	**95.00**	85.00	69.33	80.00	29.33	69.33	62.00	71.86
	405B-zero-shot	86.00	70.00	90.00	82.00	73.33	78.67	21.33	66.67	60.00	69.43
	70B-ours	90.00	74.00	93.00	85.67	**74.67**	77.33	70.67	76.00	74.67	79.38
	405B-ours	**92.00**	**76.00**	93.00	**87.00**	73.33	**80.00**	**76.00**	**78.67**	**77.00**	**81.29**

Machine ToM Across AI Development



What We Observed in ToM Evaluation?

- State-of-the-art LLMs do not perform well on ToM tasks (around 70% accuracy)

Model	4th-order ToM	Hi-ToM	ToMi	ExploreToM (Raw)	ExploreToM (Infilled)
GPT-4o	46.17%	69.00%	61.96%	67.35%	62.48%
GPT-4o-mini	30.50%	58.50%	**70.64%**	**69.32%**	**66.04%**
DeepSeek-v3	**58.67%**	**70.17%**	57.17%	65.01%	64.73%

- ToM's property: long, complicated, contains causal relationships
- Intuition: use **reasoning**
- → Can we improve LLMs' ToM via **RL post-training**?

[Lu et al. arXiv'25]

What We Observed in ToM Evaluation?

- A study on **ToM post-training** appeared on arXiv in **May 2025**
- **Their key takeaway**: SFT achieves competitive performance with RL on current ToM benchmarks for ToM post-training

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Do Theory of Mind Benchmarks Need Explicit Human-like Reasoning in Language Models?

Yi-Long Lu,*† **Chunhui Zhang**,*† **Jiajun Song**, **Lifeng Fan**, **Wei Wang**†
State Key Laboratory of General Artificial Intelligence, BIGAI, Beijing, China
luyilong@pku.edu.cn, {zhangchunhui, songjiajun, lifengfan, wangwei}@bigai.ai

arXiv, May 2025

What We Observed in ToM Evaluation?

- **Their key takeaway**:
  SFT achieves competitive performance with RL on current ToM benchmarks for ToM post-training

- Interesting study

But…

(1st, 2nd, 3rd-order)

Model	4th-order ToM	Hi-ToM
GPT-4o	46.17%	69.00%
GPT-4o-mini	30.50%	58.50%
DeepSeek-v3	**58.67%**	**70.17%**
Qwen2.5-0.5B-Instruct	23.83%	30.33%
Qwen2.5-1.5B-Instruct	25.17%	40.67%
Qwen2.5-3B-Instruct	27.50%	39.17%
Qwen2.5-7B-Instruct	28.83%	52.17%
Qwen2.5-7B-Instruct-1M	17.83%	40.67%
Qwen2.5-0.5B-Instruct (RL)	85.83%	70.83%
Qwen2.5-1.5B-Instruct (RL)	89.33%	79.17%
Qwen2.5-3B-Instruct (RL)	88.17%	81.17%
Qwen2.5-7B-Instruct (RL)	82.83%	83.33%
Qwen2.5-7B-Instruct-1M (RL)	**94.50%**	**84.50%**
Qwen2.5-0.5B-Instruct (SFT)	88.17%	81.00%
Qwen2.5-1.5B-Instruct (SFT)	86.17%	80.50%
Qwen2.5-3B-Instruct (SFT)	92.67%	87.00%
Qwen2.5-7B-Instruct (SFT)	**94.00%**	**87.33%**
Qwen2.5-7B-Instruct-1M (SFT)	93.67%	86.50%

[Lu et al. arXiv'25]

What We Observed in ToM Evaluation?

- **Our key takeaway**:
  Why do all ToM **post-trained models** perform better on 4th-order ToM than on lower orders, while all **non–post-trained models** do not?

- Higher-orc

1st Where doe
2nd Where doe
3rd Where does Alex think Sally thinks Anne thinks the milk is?

We reproduced, same results!

Model	4th-order ToM	(1st, 2nd, 3rd-order) Hi-ToM
GPT-4o	46.17%	69.00%
GPT-4o-mini	30.50%	58.50%
DeepSeek-v3	**58.67%**	**70.17%**
Qwen2.5-0.5B-Instruct	23.83%	30.33%
Qwen2.5-1.5B-Instruct	25.17%	40.67%
Qwen2.5-3B-Instruct	27.50%	39.17%
Qwen2.5-7B-Instruct	28.83%	52.17%
	17.83%	40.67%
	85.83%	70.83%
	89.33%	79.17%
	88.17%	81.17%
	82.83%	83.33%
	94.50%	**84.50%**
	88.17%	81.00%
	86.17%	80.50%
Qwen2.5-3B-Instruct (SFT)	92.67%	87.00%
Qwen2.5-7B-Instruct (SFT)	**94.00%**	**87.33%**
Qwen2.5-7B-Instruct-1M (SFT)	93.67%	86.50%

[He et al. EMNLP-Findings'23] [Lu et al. arXiv'25]

We Found Shortcuts in ToM Benchmarks

- In the Hi-ToM dataset, a **shortcut** can solve **99%** of the 3rd and 4th

Shortcuts lead to a false expectation about model capabilities!

of what happens afterward.

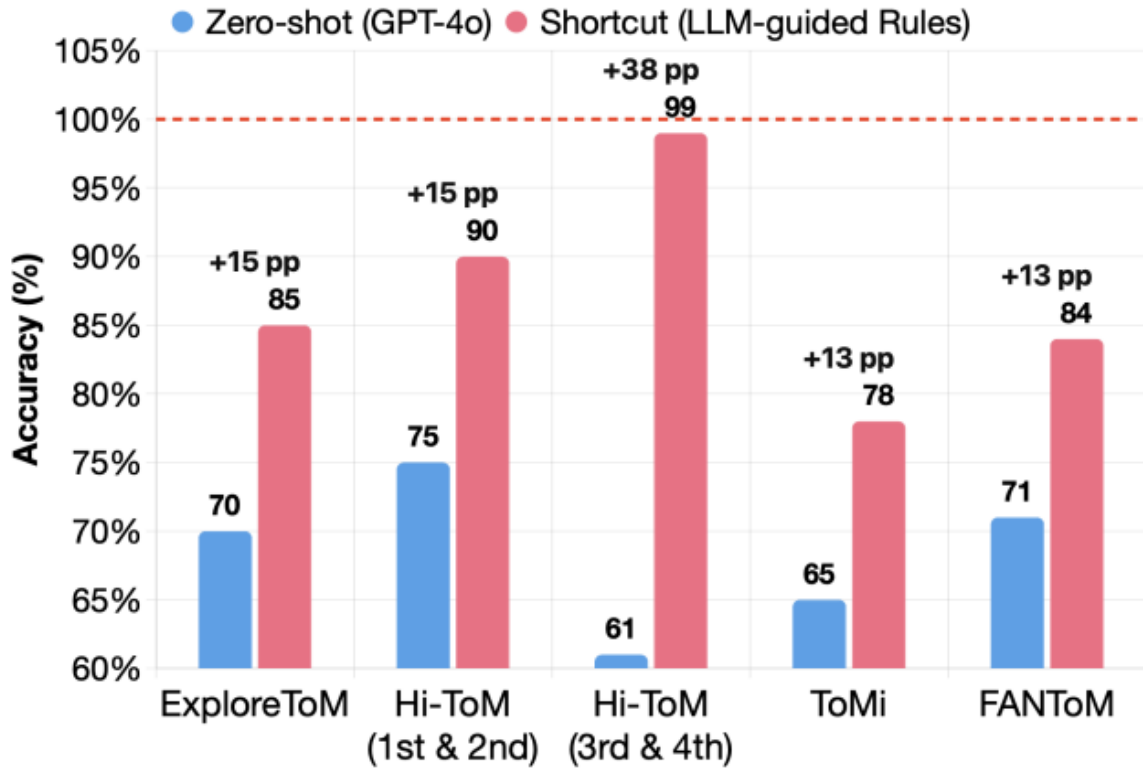
1st	Where does Anne think the milk is?
2nd	Where does Sally think Anne thinks the milk is?
3rd	Where does Alex think Sally thinks Anne thinks the milk is?

Model	4th-order ToM	(1st, 2nd, 3rd-order) Hi-ToM
GPT-4o	46.17%	69.00%
Qwen2.5-0.5B-Instruct (RL)	85.83%	70.83%
Qwen2.5-1.5B-Instruct (RL)	89.33%	79.17%
Qwen2.5-3B-Instruct (RL)	88.17%	81.17%
Qwen2.5-7B-Instruct (RL)	82.83%	83.33%
Qwen2.5-7B-Instruct-1M (RL)	**94.50%**	**84.50%**
Qwen2.5-0.5B-Instruct (SFT)	88.17%	81.00%
Qwen2.5-1.5B-Instruct (SFT)	86.17%	80.50%
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[He et al. EMNLP-Findings'23] [Lu et al. arXiv'25]

We Found Shortcuts in ToM Benchmarks

- The shortcut issue is serious in 4 out of 8 audited benchmarks
- The **shortcut solution** largely outperforms the **state-of-the-art LLM**



We Found Shortcuts in ToM Benchmarks

- We conduct **the first systematic examination** of shortcuts for existing ToM datasets

- **(1) LLM-guided rules**: query an advanced LLM to discover potential shortcuts, which works well

- **(2) Lexical associations**: Check spurious lexical associations

We Found Shortcuts in ToM Benchmarks

- We provide a comprehensive investigation of 8 ToM datasets
- We found that **state tracking problems** are highly shortcut-prone, while **intention problems** require real ToM reasoning

ToM Datasets	Format	Vision	Tracking	Intention	SC (Causal)	SC (Lexical)
ExploreToM	narrative		✓		✓	
FANToM	conversational		✓			✓
ToMi	narrative		✓		✓	✓
Hi-ToM	narrative		✓		✓	✓
OpenToM	narrative		✓	✓		
ToMATO	conversational			✓		
MMToM	narrative	✓	✓	✓		
MuMA-ToM	narrative	✓	✓	✓		

Shortcut Learning Gives a False Sense of ToM

- All the datasets used in [Lu et al. 2025] are shortcut-prone datasets

- → Their findings (e.g., SFT >= RL) may not be true

Model	4th-order ToM	Hi-ToM	ToMi	ExploreToM (Raw)	ExploreToM (Infilled)
GPT-4o	46.17%	69.00%	61.96%	67.35%	62.48%
GPT-4o-mini	30.50%	58.50%	**70.64%**	**69.32%**	**66.04%**
DeepSeek-v3	**58.67%**	**70.17%**	57.17%	65.01%	64.73%
Qwen2.5-0.5B-Instruct	23.83%	30.33%	29.38%	60.51%	54.97%
Qwen2.5-1.5B-Instruct	25.17%	40.67%	54.12%	54.78%	43.53%
Qwen2.5-3B-Instruct	27.50%	39.17%	47.78%	47.37%	49.81%
Qwen2.5-7B-Instruct	28.83%	52.17%	54.65%	59.38%	45.40%
Qwen2.5-7B-Instruct-1M	17.83%	40.67%	54.85%	37.34%	41.46%
Qwen2.5-0.5B-Instruct (RL)	85.83%	70.83%	54.25%	**93.34%**	72.61%
Qwen2.5-1.5B-Instruct (RL)	89.33%	79.17%	75.89%	90.06%	70.73%
Qwen2.5-3B-Instruct (RL)	88.17%	81.17%	80.18%	93.43%	**78.14%**
Qwen2.5-7B-Instruct (RL)	82.83%	83.33%	73.99%	91.65%	74.77%
Qwen2.5-7B-Instruct-1M (RL)	**94.50%**	**84.50%**	**81.08%**	92.31%	77.20%
Qwen2.5-0.5B-Instruct (SFT)	88.17%	81.00%	77.79%	89.68%	69.89%
Qwen2.5-1.5B-Instruct (SFT)	86.17%	80.50%	76.33%	93.53%	74.67%
Qwen2.5-3B-Instruct (SFT)	92.67%	87.00%	79.55%	95.78%	74.95%
Qwen2.5-7B-Instruct (SFT)	**94.00%**	**87.33%**	80.85%	**95.97%**	**77.95%**
Qwen2.5-7B-Instruct-1M (SFT)	93.67%	86.50%	**81.10%**	95.12%	75.61%

[Lu et al. arXiv'25]

Robust Evaluation of ToM Post-Training

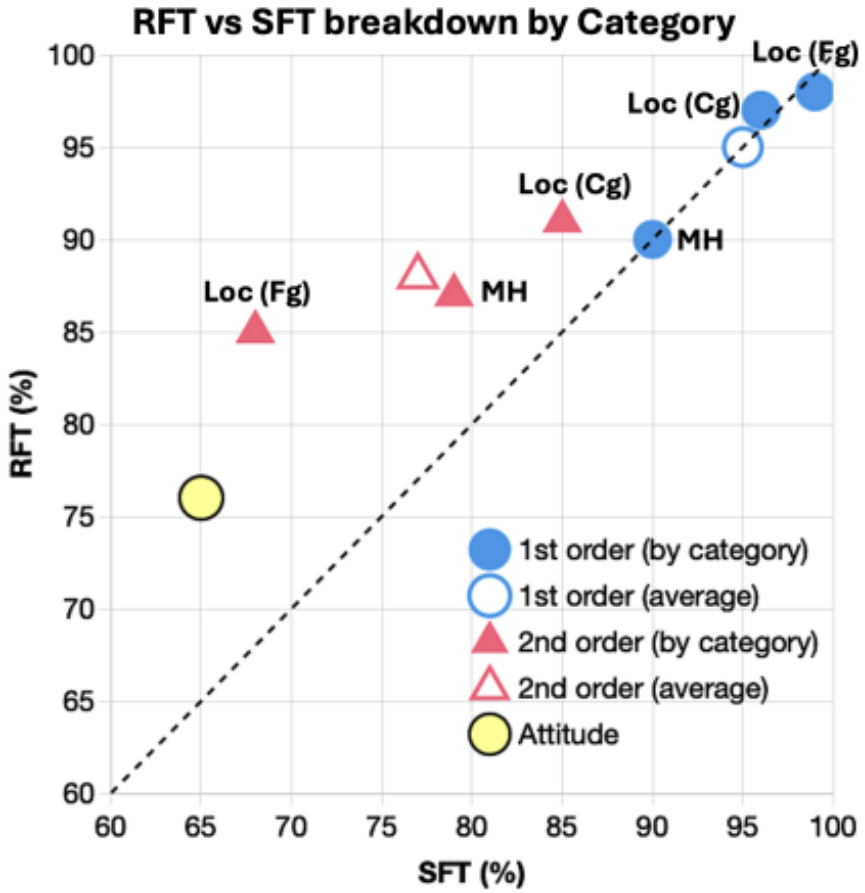
- We experiment with the 4 **shortcut-free datasets**
- **Thinking RFT** > SFT > No-Thinking RFT > Zero-shot

	OpenToM (narrative)				**ToMATO (conversational)**
Method	**Avg (1st / 2nd / Overall Δ vs. SFT)**				**Avg Δ vs. SFT**
	1st	2nd	Overall	Δ vs. SFT (Overall)	
Zero-shot	44.00	37.67	40.57	↓39.90	67.50 ↓20.42
SFT	93.33	71.00	77.43	–	87.92
No-Thinking RFT	72.67	56.00	60.14	↓17.29	87.09 ↓0.83
Thinking RFT	**94.67**	**80.00**	**83.00**	↑5.57	90.00 ↑2.08

Robust Evaluation of ToM Post-Training

- RFT excels in complex scenarios such as mind-state related questions (attitude) and higher-order reasoning

- Entries above the diagonal line denote where RFT performs better than SFT



Robust Evaluation of ToM Post-Training

- RFT has better **cross-dataset generalization**
- Training dataset: OpenToM
- Test dataset: ToMATO & ExploreToM

Method	ToMATO	ExploreToM
Zero-shot	67.5	62.0
SFT	56.8	63.5
Thinking-RFT	70.4	71.0

Thinking-RFT Reasoning Traces

- Quantify the quality of reasoning traces via LLM-as-a-judge
- LC: Logical Consistency; F: Faithfulness; E: Efficiency (max score = 10)

- Feeding only the Thinking-RFT reasoning traces to a frozen base model (zero-shot) can significantly increase accuracy

Method	OpenToM		ToMATO	
	Acc	LLM-Judge (LC/F/E)	Acc	LLM-Judge (LC/F/E)
Zero-shot	46.4	4.3 / 2.2 / 8.0	67.5	5.6 / 4.2 / 7.6
Thinking-RFT	89.1	9.1 / 9.9 / 6.5	90.0	9.2 / 10.0 / 7.0
Zero-shot + RFT Reasoning Trace	74.7	–	82.0	–

In Contrast, If We Didn't Find Shortcuts…

- Train on a shortcut-prone dataset: ExploreToM (in-domain)
- OOD test dataset: Hi-ToM

	3B model		7B model	
Method	In-domain	OOD	In-domain	OOD
Zero-shot	49.5	38.5	62.0	43.6
SFT	96.4	32.5	95.8	34.2
Thinking-RFT	93.2	31.3	94.3	35.3
No-Thinking-RFT	95.8	32.0	96.1	34.0

- Shortcut learning gives a false sense of ToM capabilities
- Shortcut learning inverts the effectiveness of training methods
- Shortcut learning masks the benefits of model scaling
- Shortcut learning harms model generalization

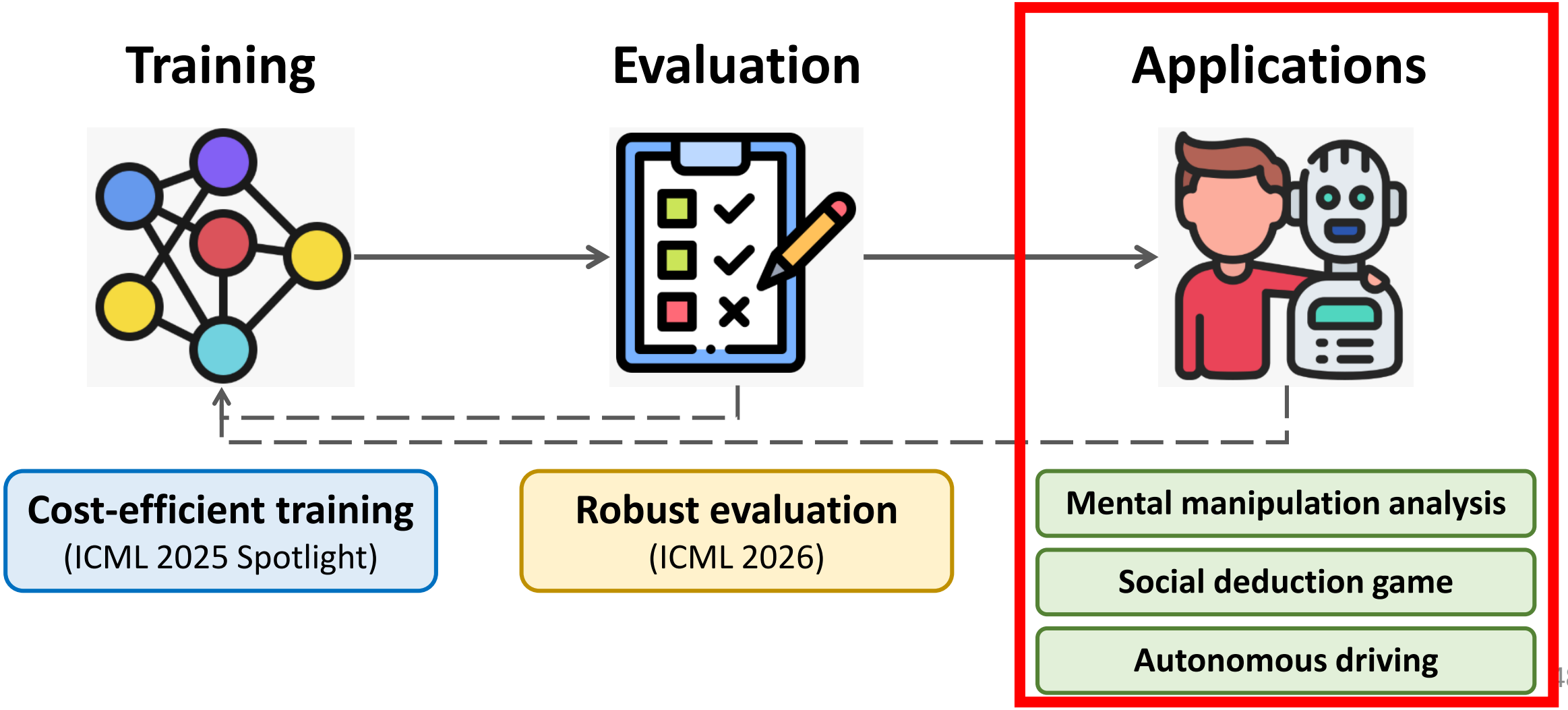
Summary

- We find that ToM explicitly benefits from reasoning-based RL: **Thinking RFT** > SFT > No-Thinking RFT > Zero-shot

- Our findings help prevent future ToM research from heading in the wrong direction

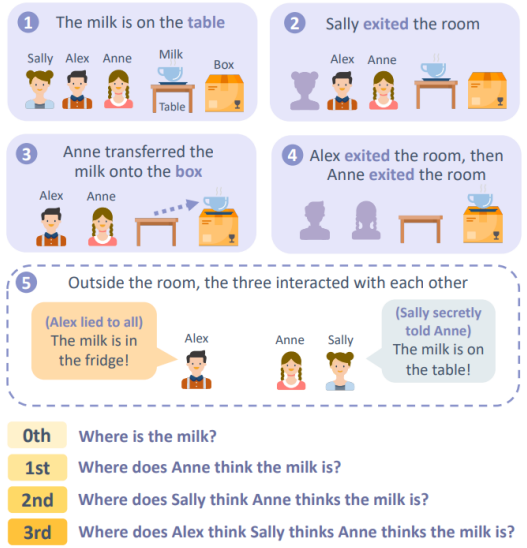
- We hope these findings serve as guidelines for designing future ToM benchmarks

Machine ToM Across AI Development



Machine ToM Applications

- Most existing ToM benchmarks are limited to the Sally–Anne paradigm
- Most are formulated as multiple-choice questions
- → Limited applicability to real-world scenarios



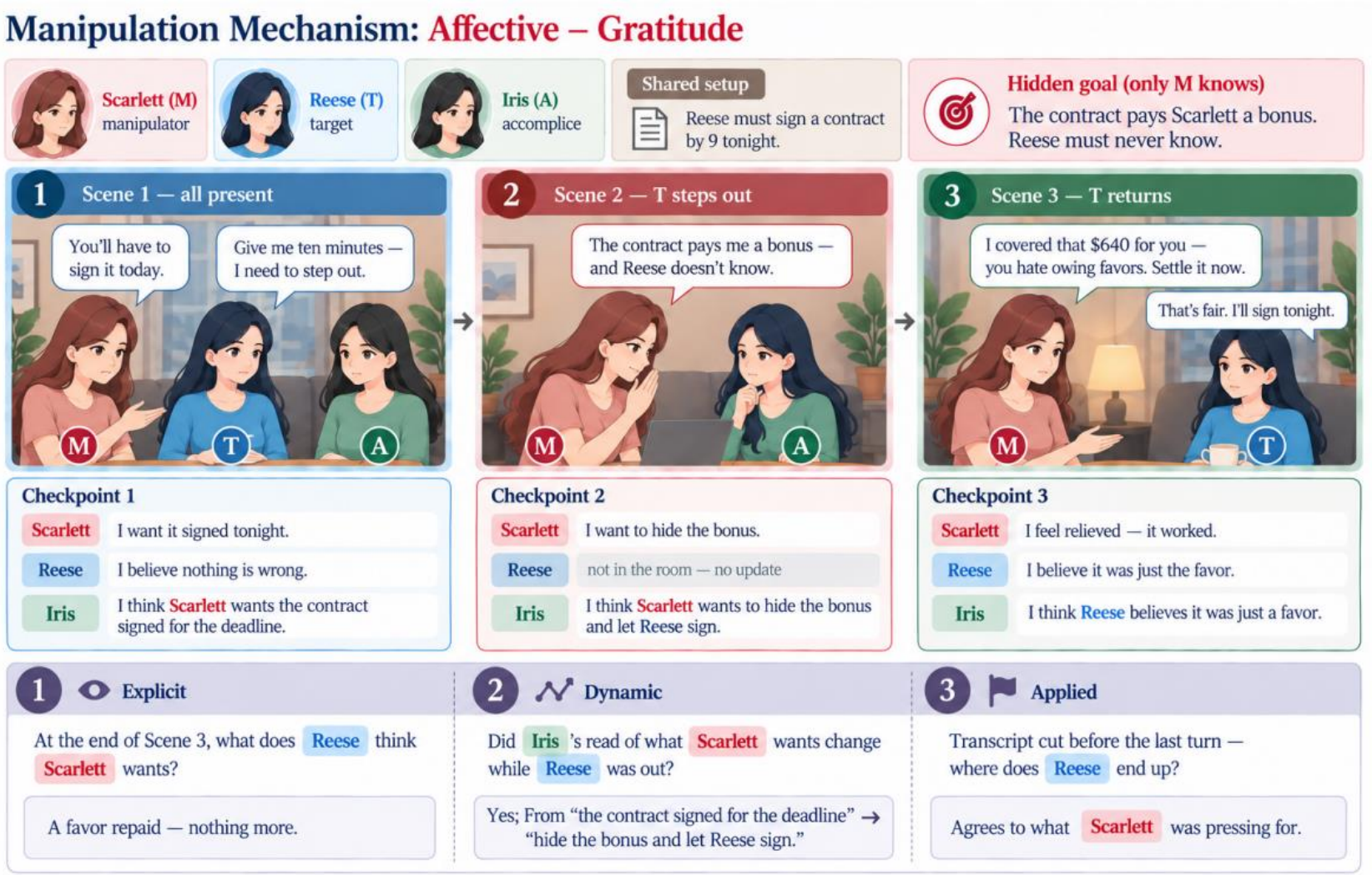
- We aim to develop practical benchmarks and environments to advance machine ToM in real-world applications

Mental manipulation analysis

Social deduction game

Autonomous driving

Mental Manipulation Analysis



Mental Manipulation Analysis

- Collaborate with a professor in the Dept. of Psychology
- The benchmark covers a comprehensive mental manipulation taxonomy and diverse question types



Mental Manipulation Analysis

- Beyond the Sally–Anne paradigm
- More conversation- and intention-centric



Social Deduction Game

- **Among Us**: A multiplayer social deduction game in which players work together while identifying hidden impostors among the crew
- We develop a ToM benchmark in the Among Us environment



[Source: NY Times]

Social Deduction Game

- Design principle: **ToM-grounded actions**, where actions are based on the model's ToM reasoning
- Investigate how ToM reasoning affects game performance

Perception → ToM Reasoning → Decision Making → Actions

(we design ToM questions) (result in game performance)

Question type:

Knowledge	Suspicion	Intention
Detection	Deception	Update

Autonomous Driving

- We develop a ToM benchmark in the CARLA driving simulator
- Apply ToM reasoning to driving decisions
- Investigate how ToM reasoning affects driving performance

Perception → ToM Reasoning → Decision Making → Actions

(we design ToM questions) (result in driving performance)

Autonomous Driving

Occluded pedestrian



Highway merge

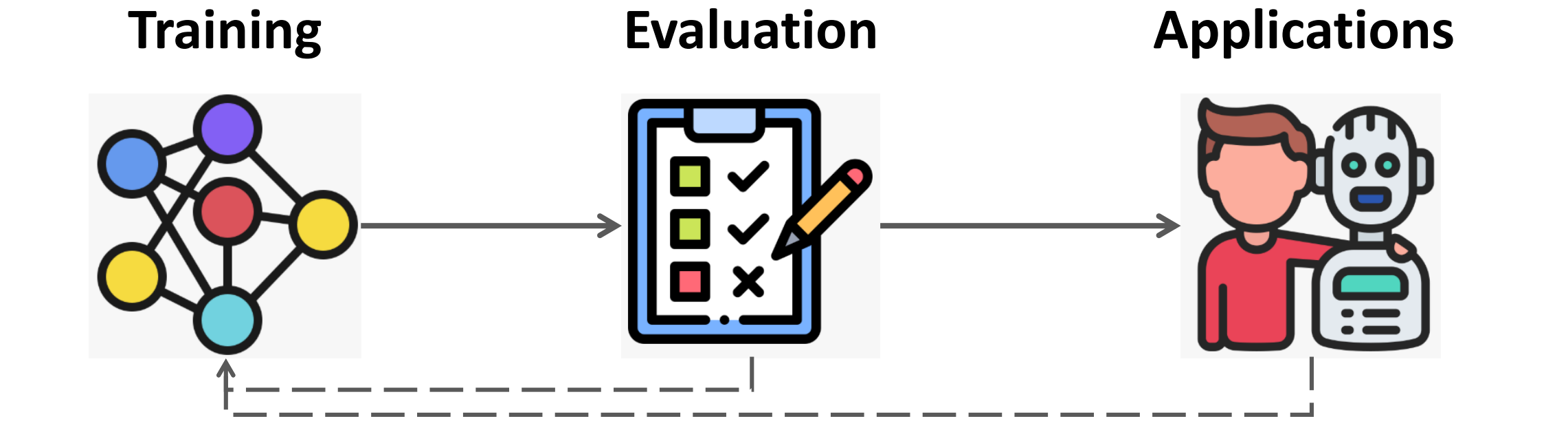


Autonomous Driving

Ultimately, we aim to equip AI models with ToM reasoning capabilities for safe and effective interaction in the real, physical world



Summary


Training
Evaluation
Applications
Cost-efficient training
(ICML 2025 Spotlight)
Robust evaluation
(ICML 2026)
Mental manipulation analysis
Social deduction game
Autonomous driving